

NANDAN N N

Electronics & Instrumentation Undergraduate

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SUMMARY

Electronics & Instrumentation Engineering undergraduate specializing in Embedded Systems, with hands-on experience across STM32, Raspberry Pi, FPGA, and Embedded C. Skilled in real-time firmware development, sensor interfacing (UART, SPI, I2C), PID-based control systems, and FPGA-based digital design using Verilog. Also experienced in web development.

EDUCATION

B.E., Electronics & Instrumentation Engineering — SJCE, JSS Science & Technology University, Mysore *Expected 2027*

Pre-University (PUC) — Ambika PU College, Puttur *2023*

TECHNICAL SKILLS

- Programming: MATLAB, C, Embedded C, Verilog HDL, Python
- Hardware Platforms: STM32, Raspberry Pi, FPGA, Arduino, ESP32
- Protocols/Interfaces: UART, I2C, SPI, GPIO, PWM
- Tools: ModelSim/Questasim, Keil, Proteus, Multisim, Xilinx
- Web: HTML, CSS, JavaScript, Bootstrap, jQuery, React

PROJECTS

RTL Design & Verification of RAM

Verilog HDL, Questasim

- Designed and verified a synchronous RAM at the RTL level using Verilog HDL, implementing memory read/write operations with a synchronous clock.
- Developed a functional testbench and validated memory functionality through simulation, read/write operations, and waveform analysis.

RTL Design & Verification of ROM

Verilog HDL, Questasim

- Designed and verified a synchronous ROM at the RTL level using Verilog HDL, implementing address-based data read operations.
- Developed a functional testbench and validated address decoding and data retrieval through simulation and waveform analysis.

RTL Design & Verification of Synchronous FIFO

Verilog HDL, Questasim

- Designed and verified a synchronous FIFO at the RTL level using Verilog HDL, implementing data write and read operations using a common clock.
- Developed a functional testbench and validated FIFO operation, including full and empty conditions, through simulation and waveform analysis.

RTL Design & Verification of Asynchronous FIFO

Verilog HDL, Questasim

- Designed and verified an asynchronous FIFO at the RTL level using Verilog HDL, enabling data transfer between independent read and write clock domains.
- Developed a functional testbench and validated FIFO operation, including full and empty conditions, through simulation and waveform analysis.

Smart Self-Balancing Vehicle — STM32 (Embedded Control System)

STM32 Blue Pill, MPU6050, Embedded C, PID Control, UART, I2C

- Engineered a two-wheel self-balancing robot using STM32 Blue Pill and MPU6050 IMU.
- Formulated and tuned a PID-based real-time stabilization algorithm for dynamic recovery.
- Integrated I2C sensor acquisition and PWM motor actuation with calibration, noise filtering, and UART-based debugging.

Adaptive Eating Assistance System

ESP32, Servo Motors, MPU6050, Embedded C

- Designed an ESP32-based assistive robotic feeding system integrating MPU6050 IMU and ultrasonic sensor for real-time spoon stabilization and object detection.
- Developed Embedded C firmware for servo motor control, I2C sensor interfacing, and real-time feedback processing.

Smart Pothole and Hump Detection System

Raspberry Pi, Python, Ultrasonic Sensors

- Built a Raspberry Pi-based system for real-time pothole and speed-breaker detection using ultrasonic sensors.
- Implemented Python-based sensor interfacing, distance measurement, GPIO control, and obstacle detection logic to enable adaptive vehicle speed control.

TECHNICAL DEVELOPMENT

RTL and Design Verification Course — KT Semicon	2026
Software Engineer Intern — 1Xsoft Private Limited	2026 (3 months)
Embedded Systems Intern — InternPE	2026 (1 month)
RISC-V FPGA IP Development Intern — VSDFPGA	2026 (15 days)

ACHIEVEMENTS

- Represented college at the Suttur Jatra Science & Technology Exhibition, presenting a technical project and interacting with participants and evaluators.
- Social Media Lead, College Music Club — led digital promotions and coordinated events in team-driven environments.

DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief.